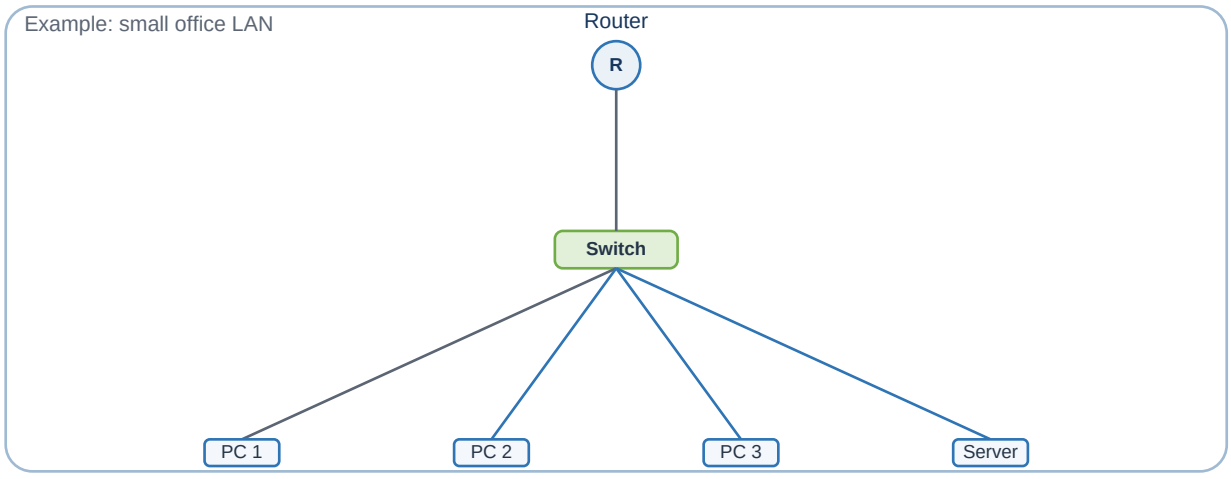
CCNA LEARNING SERIES

Day 1: Networking Fundamentals

Concepts • Devices • IP basics • Router configuration

Learning objective
Understand computer networks, identify common networking devices, learn basic network terms and configure an IP address on a Cisco router interface.

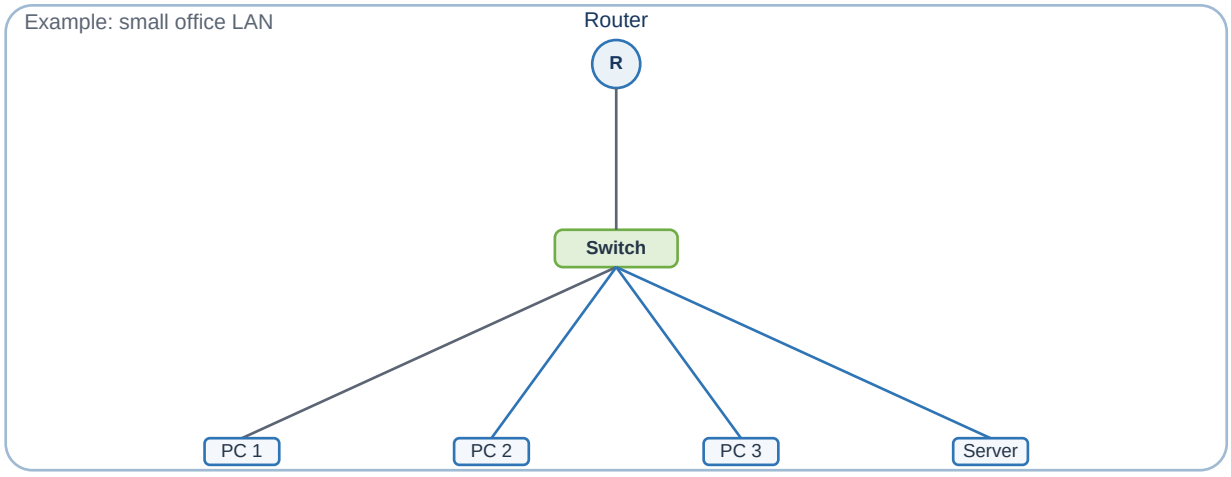


Based on handwritten study notes and expanded with beginner-friendly explanations and examples.

1. What is a Network?

A **computer network** is a group of connected devices that communicate with one another and share data and resources. Devices can include computers, servers, printers, smartphones, switches and routers.

Example: In an office, several computers connect to a switch. The switch connects to a router, and the router connects the office network to other networks or the internet.



Why do we use networks?

- To share files and information.
- To share resources such as printers and internet access.
- To communicate between devices.
- To access services hosted on servers.

2. Transmission Media: Guided and Unguided

Transmission media is the path through which data travels from one device to another.

Type	Meaning	Examples
Guided media	Data travels through a physical or wired path.	Ethernet twisted-pair cable, coaxial cable, fibre-optic cable.
Unguided media	Data travels wirelessly through the air using electromagnetic waves.	Wi-Fi, Bluetooth and cellular/mobile communication.

3. Basic Network Devices

Router

A **router** connects different IP networks and forwards packets between them. A home router commonly connects your local network to the internet. Routers make forwarding decisions using routing information.

Switch

A **switch** connects devices within a LAN. A typical Layer 2 switch learns MAC addresses and forwards Ethernet frames toward the correct port.

Hub

A **hub** is a basic Layer 1 device. It repeats incoming signals out of its other ports. It does not learn MAC addresses or intelligently choose the destination port. Hubs are largely obsolete in modern networks.

Remember: A switch mainly connects devices inside a LAN. A router connects different networks. A hub repeats signals to its other ports.

4. Types of Networks

Type	Full form	Typical coverage	Example
PAN	Personal Area Network	Around one person	Phone connected to wireless earbuds by Bluetooth
LAN	Local Area Network	Room, home, office or building	Computers connected in an office
MAN	Metropolitan Area Network	City or large town	Network connecting sites across a city
WAN	Wide Area Network	Large geographic areas	Company network connecting offices in different cities

5. IP Address and MAC Address

IP address

An **IP address** is a logical address assigned to a network interface so it can communicate over an IP network. An IPv4 address is 32 bits and is usually written as four decimal numbers separated by dots.

Example IPv4 address: 192.168.1.10
Example subnet mask: 255.255.255.0

MAC address

A **MAC address** is a link-layer address used on a local Ethernet or Wi-Fi network. It is commonly written as six pairs of hexadecimal characters, for example `00:1A:2B:3C:4D:5E`. A device may have a different MAC address for each network interface, and some devices support MAC randomization.

IP address	MAC address
Logical address used for communication across IP networks.	Link-layer address used for local network delivery.
Can be assigned manually or automatically, for example by DHCP.	Associated with a network interface; it may be changed or randomized by software.

6. Important Network Terms

DHCP – Dynamic Host Configuration Protocol

DHCP automatically provides network settings to client devices. These settings can include an IP address, subnet mask, default gateway and DNS server. This reduces the need to configure each device manually.

DNS – Domain Name System

DNS translates human-readable domain names, such as `www.example.com`, into IP addresses that computers use to locate services.

Default gateway

The **default gateway** is the router or Layer 3 device a host sends traffic to when the destination is outside its local subnet. For example, a host with address 192.168.1.10/24 may use 192.168.1.1 as its default gateway.

Subnet mask

A **subnet mask** identifies which bits of an IPv4 address represent the network portion and which represent the host portion. For example, 255.255.255.0 is also written as /24.

Example: If a PC is configured as 192.168.1.10 with mask 255.255.255.0, it is in the 192.168.1.0/24 subnet. A common gateway address for this example network is 192.168.1.1.

7. Cisco Router and Switch Models

Your notes mention a Cisco ISR 4331 router and a Catalyst 2960 switch.

• **Cisco ISR 4331:** an Integrated Services Router model used for routing and other branch-office network services. Exact interface names and features depend on the installed hardware and software.
• **Cisco Catalyst 2960:** a family of Layer 2 access switches commonly used to connect end devices in a LAN. Port count and features depend on the exact model.

Why use a switch with a router?

A router may have a limited number of built-in LAN interfaces for the required design. A switch provides multiple access ports so many PCs, printers and servers can connect to the same LAN. The switch then connects to the router.

8. Basic Cisco Router Interface Configuration

The following example assigns 192.168.1.1/24 to a router interface. Use the actual interface name shown by your router; on an ISR 4331, a common name is `GigabitEthernet0/0/0`.

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# end
Router# show ip interface brief
```

What each command does

Command	Purpose
enable	Moves from user EXEC mode to privileged EXEC mode.
configure terminal	Enters global configuration mode.
interface gigabitEthernet 0/0/0	Selects the router interface to configure.
ip address 192.168.1.1 255.255.255.0	Assigns the IPv4 address and subnet mask to that interface.
no shutdown	Administratively enables the interface.
show ip interface brief	Displays a summary of interface IP addresses and status.

Important lab note: The interface must exist on your router, and the connected cable/device must be correct. The interface may show `up/up` only when physical and line-protocol conditions are working. Check the actual interface name and status before troubleshooting.

9. Quick Revision

• Network: connected devices that communicate and share resources.
• Router: connects different IP networks.
• Switch: connects devices within a LAN and forwards frames using MAC addresses.
• Hub: repeats signals to its other ports.
• DHCP: automatically supplies IP configuration.
• DNS: resolves domain names to IP addresses.
• Default gateway: next hop for traffic leaving the local subnet.
• Subnet mask: separates network and host portions of an IPv4 address.

10. Interview Practice Questions

Q1. What is a computer network?
A group of connected devices that communicate and share data or resources.

Q2. What is the main difference between a router and a switch?
A switch connects devices within a LAN, while a router forwards traffic between different IP networks.

Q3. What is DHCP used for?
It automatically provides IP configuration, such as an IP address, subnet mask, gateway and DNS server.

Q4. What is DNS?
DNS translates domain names into IP addresses.

Q5. Why do we configure `no shutdown`?
It removes the administrative shutdown state and enables the interface.

End of Day 1 | Next topic: OSI Model and TCP/IP Model

Study note: This document is based on handwritten notes and expanded for learning. Interface names, available ports and command support vary by device model and IOS version. Verify commands in your own lab.